

GulfMart Supply Chain Intelligence

Executive Report: findings, risks and evidence-based recommendations

Prepared for: GulfMart Retail Group executive team (COO, Supply Chain, Procurement, Finance)

Scope: sales, inventory, suppliers and logistics, 01 Jan 2015 to 03 Sep 2018

Basis: the GulfMart Supply Chain Intelligence Platform (Phases 1–10)

Read this first. GulfMart is a simulated client. Transactions come from public datasets (Olist, DataCo) relabelled into a UAE context. Inventory, purchase-order, return and cost data are **simulated** from distributions of the real data. Every figure below is labelled as observed (public data), modelled (simulated) or derived. Recommendations describe what the evidence supports *within this model*. They are not claims about a real retailer.

1. Executive summary

GulfMart's network generated **AED 148.55M** in revenue from **496,729** units over the analysis period, at a gross margin of **AED 41.33M** (27.8%). The platform now gives management one governed view of sales, inventory, suppliers and logistics, and the evidence points to one dominant operational problem: **replenishment does not account for lead time, so availability fails where supply is slowest.**

Headline KPI	Value	What it says
Stockout rate (modelled)	13.5% of product-days	Roughly one day in seven, a tracked product is out of stock
Fill rate (modelled)	90.2%	40,231 of 410,705 units demanded could not be served
Supplier on-time delivery (observed)	42.7%	Fewer than half of shipments arrive by the promised date
Average lead time (observed)	3.5 days	Short, but unreliable: the lateness is systemic, not supplier-specific
Inventory value, latest (modelled)	AED 2.83M	Capital currently held in the top-502 tracked products
Inventory turnover (modelled)	121.8×	Lean stock: little buffer against disruption

Key findings

1. **Stockouts are driven by lead time.** Products with 11–14 day replenishment lead times are out of stock on 20.9% of days, against 8.2% for 3–7 day products (Spearman $p = 0.84$ across 502 products).
2. **Lead time is the biggest risk lever; transport cost is not.** In the scenario engine, a 50% lead-time increase lifts stockout probability from 13.9% to 88.6%. A 40% transport-cost increase changes cost but not availability.
3. **Late delivery is network-wide.** On-time delivery is 42.7%, and lead times do not differ significantly between suppliers (Kruskal-Wallis $p = 0.139$). Switching suppliers would not fix it.
4. **Only one seasonal window moves demand.** Back-to-school raises daily revenue significantly ($p = 0.011$). Ramadan, Eid, National Day, Summer and Year-end do not (all $p > 0.4$).

Top recommendations (evidence and trade-offs in \$10)

#	Recommendation	Expected effect (modelled)
R1	Set reorder points at <i>lead time + 3 days</i> of demand instead of a flat 7 days	Stockout 13.5% → 8.4%; fill 90.2% → 97.6%; +30,216 units served; inventory +28.7%
R2	Invest in lead-time reliability before transport-cost savings	Lead time –30% cuts stockout probability to 0.09%
R3	Pre-build stock for Back-to-school only, not for Ramadan/Eid/National Day	Targets the only statistically significant seasonal uplift
R4	Plan safety stock on Seasonal Naive intervals, not XGBoost's	Seasonal Naive intervals capture 77% of actuals vs 47%
R5	Put the 2 High-risk suppliers on corrective review, targeting cancellations and cost volatility	Addresses the specific drivers of their scores

2. Business problem

GulfMart operates across Dubai, Abu Dhabi, Sharjah, Ajman and Ras Al Khaimah. Before this platform, decisions were made from disconnected exports, and no one could answer basic cross-functional questions with evidence. The engagement set out seven questions (`docs/business_requirements.md` §2):

#	Question	Answered in
1	What products and regions are driving revenue?	§5 Sales findings
2	Where are we experiencing stockouts and excess inventory?	§6 Inventory findings
3	Which suppliers are creating operational risk?	§7 Supplier & logistics findings
4	Can we forecast future demand?	§8 Forecasting
5	What unusual transactions or inventory movements need investigation?	§9 Anomaly detection
6	What happens if demand, lead time or transport cost change?	§9 Scenario analysis
7	Which decisions improve service levels while controlling cost?	§10 Recommendations

Each stakeholder has a primary Power BI page. The COO and Finance use **Executive Overview** and **Scenario Simulator**. The Supply Chain Manager uses **Inventory**, and Procurement uses **Supplier & Logistics**. Every KPI in this report is defined once, in `docs/kpi_dictionary.md`, and computed identically in SQL, Python, DAX and Excel.

3. Data architecture

Sources. Two public transactional datasets carry the observed data: Olist (Brazilian e-commerce; 99,441 orders, 112,650 order lines) and DataCo (global supply chain; 180,519 order lines with shipping dates and delivery status). UAE Open Data trade statistics (21,593 rows) were profiled and staged for context. Planned M5 and UAE CPI sources were not loaded, and nothing in this report depends on them. The two transactional sources cover different periods: DataCo runs to Jan 2018 and Olist starts in Sep 2016 (Figure 1).

Pipeline. A Python ETL (`etl/`) extracts, validates, standardises currency to AED, relabels source geography into the five emirates, and loads PostgreSQL through `staging` into a `warehouse` star schema. That schema has **5 fact tables** (sales 293,169 rows, shipments 180,519, inventory 181,903, purchase orders, returns) and **8 dimensions** (date, product, customer, location, store, supplier, transport, warehouse). Simulated tables are generated deterministically (`etl/synthesize/`), so every run reproduces the same data. An Airflow DAG orchestrates the pipeline.

Analytics and delivery. On top of the warehouse sit 39 SQL analysis queries, Python modules for EDA, forecasting, supplier risk, anomaly detection and scenario modelling, a FastAPI service (`/api/v1/kpis`, `/api/v1/scenario/simulate`), a Power BI project (5 pages, 44 DAX measures), and two Excel workbooks. The workbooks independently recompute the headline KPIs and replay the scenario engine. The API and the Excel cross-check run off the same ground-truth module (`src/kpi/summary.py`).

Label	Meaning	Examples in this report
Observed (PUBLIC)	Real public data, cleaned but not fabricated	Revenue, units, lead times, on-time delivery
Modelled (SYNTHETIC)	Simulated from real-data distributions	Stock levels, stockouts, fill rate, unit cost, POs, returns
Derived	Computed from the above	KPIs, risk scores, forecasts, anomalies, scenarios

4. Data quality

The unified sales dataset (293,169 rows) passed validation with **no critical issues**. There is 8.38% overall missingness, concentrated in optional descriptive fields, and 4,866 outlier warnings on line-item sales value (IQR and $|z| > 3$). Those outliers are genuine high-value orders, so they were kept rather than removed. Profiling the raw sources found zero duplicate rows. The UAE trade file's 19.6% missingness comes entirely from three always-empty metadata columns.

A KPI defect was found and fixed during this phase. The original Fill Rate formula estimated unmet demand from stock movements ($\text{opening} - \text{closing} - \text{received}$). That expression reduces to $\text{sold} - 2 \times \text{received}$, which pushes the result toward 50% regardless of actual service. The platform reported **51.1%**. The Excel cross-check had "matched" it only because it reused the same formula. The corrected definition divides units fulfilled by units actually demanded (real daily sales for the same product and day), giving **90.2%**. It is now implemented identically in SQL, Python, DAX and Excel, with a regression test comparing Python against the reference SQL. *Lesson: a cross-check only validates a KPI when it re-derives the value from the written definition, not from another implementation.*

5. Sales findings

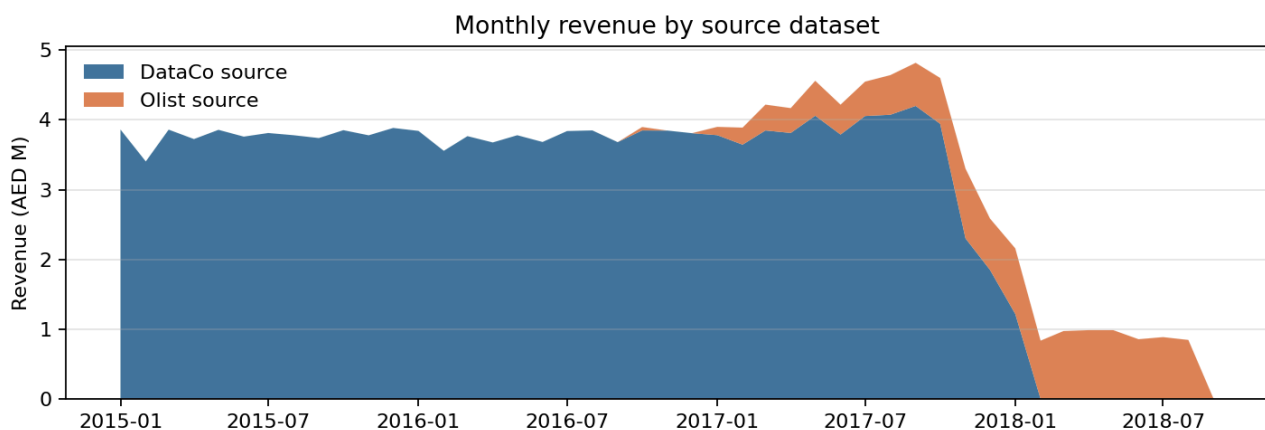


Figure 1. Monthly revenue (observed), stacked by source dataset. The step changes are source coverage, not market movements.

Growth cannot be read from this history. Revenue was AED 45.24M in 2016 and AED 49.46M in 2017 (9.3%). However, Figure 1 shows the total is two datasets with different coverage windows: Olist ramps up from Sep 2016 and DataCo ends in Jan 2018. The year-on-year change mostly reflects which source was active, so this report makes **no growth claim**. The Power BI *Revenue YoY %* measure should be read with the same caveat.

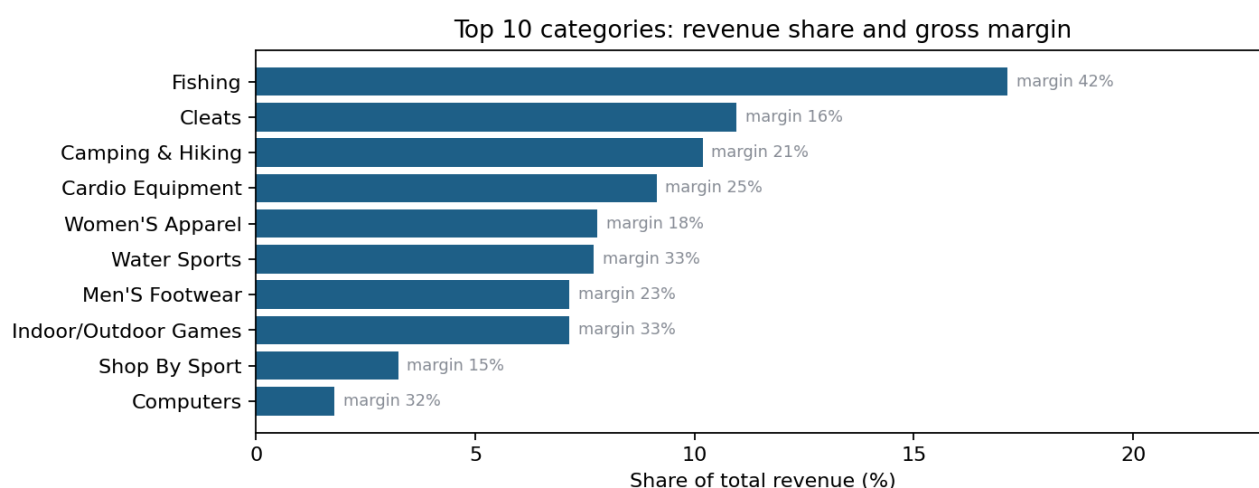


Figure 2. Revenue share (observed) and gross margin (modelled unit cost) for the ten largest of 119 categories.

Revenue is concentrated. The top three categories generate 38.3% of revenue, led by Fishing (17.1%). Margin differences between categories (Fishing 42.5%, Cleats 16.3%, Camping & Hiking 21.0%) come from **simulated** unit costs, drawn at a per-category margin rate between 15% and 45%. They illustrate the reporting capability but **are not used as evidence** for any recommendation.

Regional split is not evidence either. Abu Dhabi shows 61% of revenue, but source locations are hashed into emirates with fixed weights (`et1/transform/master_data.py`). The regional mix reflects that assignment, not UAE demand.

Seasonality was tested, not assumed. Daily revenue in each UAE seasonal window was compared with all other days (Mann-Whitney U, $\alpha = 0.05$):

Window	Days	p-value	Effect (rank-biserial r)	Result
Back-to-school	112	0.011	0.14, higher	Significant uplift
National Day	9	0.42	0.15, lower	Not significant
Ramadan	118	0.49	0.04, lower	Not significant
Summer	412	0.52	0.02, lower	Not significant
Year-end	51	0.74	0.03, higher	Not significant
Eid	24	0.75	0.04, lower	Not significant

Back-to-school averages AED 119.2k of revenue per day against AED 113.3k on regular days (+5.2%). The null results for Ramadan and Eid are valid findings in their own right: this demand history does not support holiday-specific stock builds. Discounting has no meaningful relationship with order value (Spearman $r = -0.015$). With this many rows, that coefficient is statistically non-zero but commercially negligible.

6. Inventory findings

Inventory is modelled for the 502 highest-volume products. Daily demand is the real sales history, while stock levels come from a simulated (s, S) reorder policy: reorder when stock falls to 7 days of average demand, then order up to 30 days. Each product has a replenishment lead time of 3–14 days.

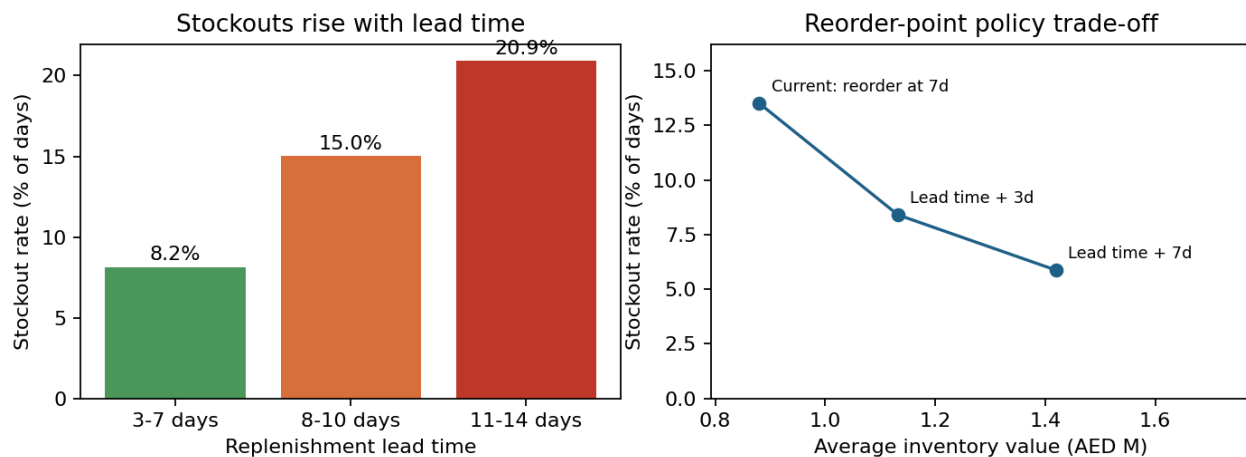


Figure 3. Left: stockout rate by replenishment lead time. Right: the same simulator re-run under three reorder policies (evidence/reorder_policy_simulation.csv).

The stockout problem is structural. A flat 7-day reorder point cannot cover a lead time longer than 7 days, so every product with slower supply runs dry before its replenishment lands:

Lead-time band	Products	Stockout rate
3–7 days	229	8.2%
8–10 days	112	15.0%
11–14 days	161	20.9%

The rank correlation between lead time and stockout rate is $\rho = 0.84$. Network-wide, 13.5% of product-days end at zero stock, and 40,231 demanded units went unserved (fill rate 90.2%). Excess stock is the smaller issue: 13 of 502 products hold more than 45 days of supply. Warehouse stockout rates range narrowly (10%–16%), which points to a policy cause rather than a site cause.

7. Supplier and logistics findings

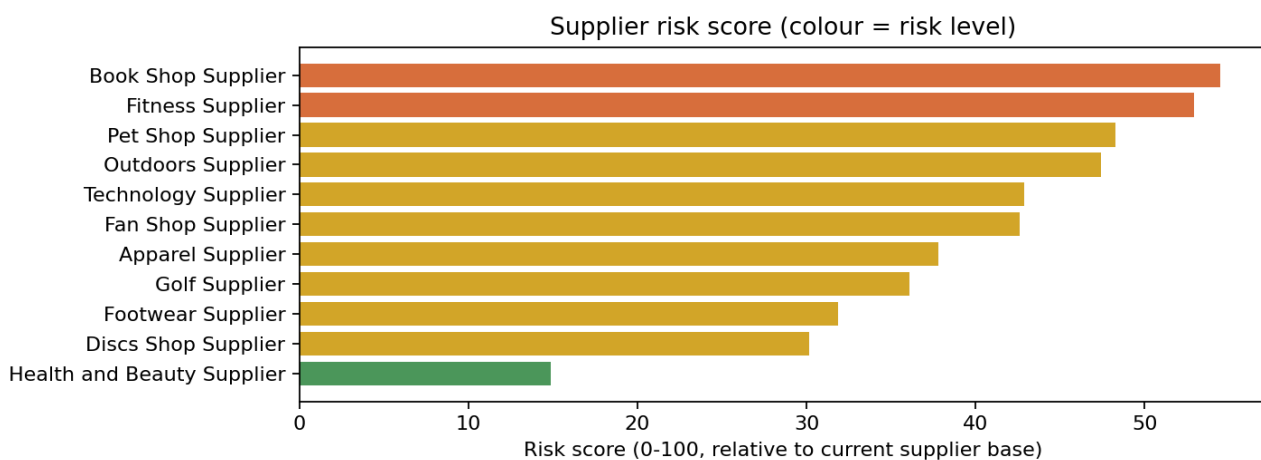


Figure 4. Supplier risk score (derived): a weighted, min-max-normalised index of late delivery 25%, defect rate 20%, lead-time variability 15%, cost volatility 15%, cancellations 15% and average lead time 10%.

Delivery performance is weak everywhere. Only 42.7% of shipments arrive by the expected date (observed). Supplier on-time rates cluster between 38.6% and 43.3%. A Kruskal-Wallis test finds **no significant difference in lead time between the 11 suppliers** ($H = 14.8$, $p = 0.139$, $\epsilon^2 \approx 0$). The lateness is a property of the network (shipping modes and carriers), not of particular suppliers.

Risk scores separate suppliers on the simulated dimensions. Book Shop Supplier (54.5) and Fitness Supplier (52.9) are rated **High**. Health and Beauty Supplier is the only Low-risk supplier (14.9). Their delivery metrics are near-identical to the rest, so the scores are driven by the procurement side:

Supplier	Cancellation rate	Cost variability (CV)	Defect proxy	Purchase orders
Book Shop Supplier	12.5% (vs ~5% typical)	0.00	2.2%	16
Fitness Supplier	4.8%	1.30	2.2%	249
Health and Beauty Supplier	0.0%	0.00	0.3%	15

Book Shop's score rests on only 16 purchase orders, so it should be confirmed with more history before any contractual action. Scores are *relative* to the current supplier base (min-max normalised), so "High" means worst-in-group, not an absolute threshold. No machine-learning benchmark was trained: no independently observed supplier-failure outcome exists, and a label built from the scoring inputs would be circular.

Logistics cost follows mode. Standard Class carries 60% of shipments at AED 51 average cost; Same Day costs AED 165 per shipment. Transport cost (modelled from distance $\times$ rate) is consistent across emirates (AED 74–75 per shipment).

8. Demand forecasting

Four models were compared on weekly demand for the 8 highest-volume products with at least 100 weeks of history. The split is strictly time-based, and every model produces an 80% prediction interval.

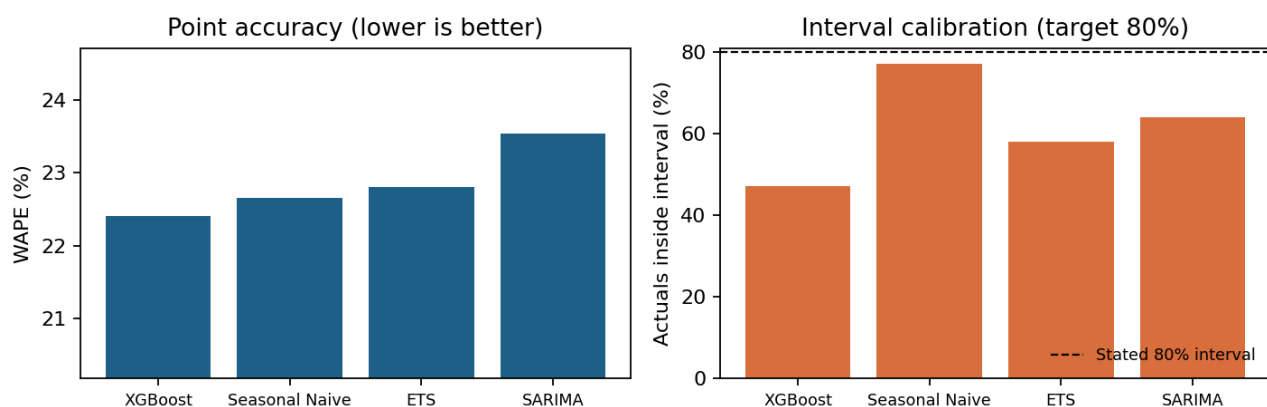


Figure 5. Mean WAPE and 80%-interval coverage across the 8 products (notebooks/06_demand_forecasting.ipynb §4).

No model is materially more accurate. WAPE ranges from 22.41% (XGBoost) to 23.54%, a spread of about one percentage point, and XGBoost's lead over Seasonal Naive (22.65%) is 0.24 points. **Calibration differs sharply**, and it matters more for stock planning. Seasonal Naive's intervals contain 77% of actual weeks, close to the stated 80%. XGBoost's contain only 47%, and ETS (58%) and SARIMA (64%) are also overconfident. Safety stock sized on an interval that is too narrow under-buffers exactly when demand

surprises. MAPE (over 250% for all models) is distorted by near-zero weeks, so WAPE is the accuracy measure used here. SARIMA and ETS logged convergence warnings with roughly 2.6 annual cycles of history, which is expected at this data volume.

9. Anomaly detection and scenario analysis

Anomaly detection

Three methods (IQR, Z-score, Isolation Forest) were run across five domains. They flagged 72,698 points in total, of which **5,292 are High severity**: 4,609 transport-cost, 622 inventory-change and 61 daily-sales anomalies (`reports/anomaly_detection/anomalies.csv`). The highest-scoring cases are inventory-change spikes at individual warehouses, 20+ times the typical daily movement (e.g. Abu Dhabi Distribution Center, 18 Jan 2015: +1,749 units vs -52 expected). Their size and direction are consistent with large replenishment receipts under the policy in §6, which a reviewer should confirm before treating them as errors.

Two method behaviours matter for any production use. IQR flags about 20% of shipment delays, because whole-day delays have a very tight core distribution. Isolation Forest always flags a fixed share, even where threshold methods correctly find nothing (supplier lead times span only 0–6 days). **Thresholds must be tuned per domain before these become alerts**, or the operations team will be flooded.

Scenario analysis

The scenario engine calibrates its safety stock so that the no-change case reproduces the observed baseline stockout rate. All values below cover the 90-day baseline window (`reports/scenario_analysis/scenario_comparison.csv`).

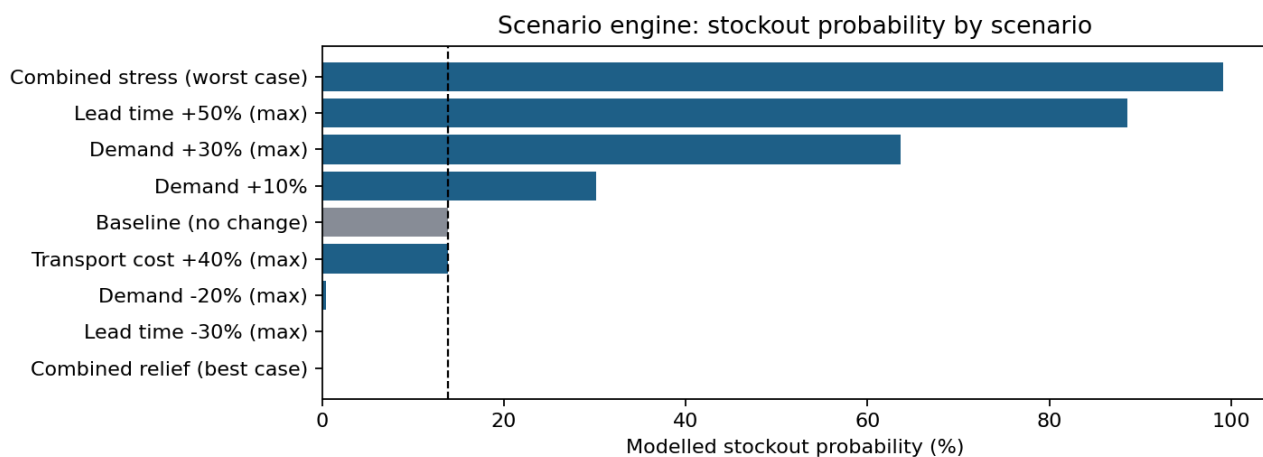


Figure 6. Modelled stockout probability by scenario; the dashed line is the calibrated baseline.

Scenario	Stockout probability	Revenue at risk (90 days)
Baseline	13.9%	AED 344k
Demand +10%	30.1%	AED 821k
Lead time +50%	88.6%	AED 2.19M
Lead time -30%	0.09%	near zero
Transport cost +40%	13.9% (unchanged)	AED 344k; transport cost AED 461k → AED 645k
Combined stress (+30% demand, +50% lead time, +40% cost)	99.1%	AED 3.19M

The buffer is thin: a modest 10% demand increase roughly doubles stockout probability. Lead time is the strongest lever in both directions. Transport-cost changes affect cost only, never availability.

10. Recommendations

Each recommendation names the metric or test behind it. Effects are *modelled* on this platform's data (see §11).

R1: Make reorder points lead-time aware. *Owner: Supply Chain Manager.* Replace the flat 7-day reorder point with *lead time + 3 days* of average demand (order-up-to raised correspondingly).

- *Evidence:* stockout rate rises from 8.2% to 20.9% across lead-time bands ($p = 0.84$, §6). Re-running the same simulator on the same real demand (`evidence/reorder_policy_simulation.csv`) gives:

Policy	Stockout rate	Fill rate	Avg inventory value
Current: reorder at 7 days	13.5%	90.2%	AED 880k
Lead time + 3 days	8.4%	97.6%	AED 1.13M (+28.7%)
Lead time + 7 days	5.9%	98.2%	AED 1.42M (+61.4%)

- *Trade-off:* the +3-day policy serves 30,216 more units (about AED 11.35M of revenue at list price) for AED 252k more average stock. Going to +7 days roughly doubles the inventory increase but serves only 2,834 further units. +3 days is the efficient point.

R2: Prioritise lead-time reliability over transport-cost savings. *Owner: COO, Procurement.* Target the delivery process (carrier SLAs, mode mix, order cut-offs) before negotiating transport rates.

- *Evidence:* lead time -30% cuts modelled stockout probability from 13.9% to 0.09%, while +50% raises it to 88.6%. A 40% transport-cost change moves cost only (AED 461k → AED 645k per 90 days) with zero availability impact (§9). On-time delivery is 42.7%, with no significant difference between suppliers (Kruskal-Wallis $p = 0.139$, §7). The fix is network-level, not a supplier switch.

R3: Pre-build stock for Back-to-school only. *Owner: Supply Chain Manager.*

- *Evidence:* Back-to-school is the only window with a significant uplift ($p = 0.011$, +5.2% daily revenue). Ramadan ($p = 0.49$), Eid ($p = 0.75$) and National Day ($p = 0.42$) show none (§5). Holiday-specific builds would add stock without a demonstrated demand response.

R4: Size safety stock on calibrated forecast intervals. *Owner: Demand Planning.* Use Seasonal Naive's intervals for safety-stock sizing. Use XGBoost for point forecasts only after recalibrating its intervals.

- *Evidence:* the models' WAPE sits within about one point of each other. Seasonal Naive's 80% intervals cover 77% of actuals, against 47% for XGBoost (§8). Interval quality, not the 0.24-point accuracy gap, decides buffer adequacy.

R5: Corrective review for the 2 High-risk suppliers, targeted at their actual drivers. *Owner: Procurement.*

- *Evidence:* Book Shop Supplier (54.5) has a 12.5% cancellation rate against about 5% for peers, but on only 16 POs, so confirm with more history first. Fitness Supplier (52.9) has cost variability of 1.30, the second highest (§7). Delivery metrics do not separate suppliers, so reviews should address cancellations and pricing stability, not delivery.

R6: Tune anomaly thresholds per domain before alerting. *Owner: Data team.*

- *Evidence:* IQR flags about 20% of shipment delays and Isolation Forest flags a fixed share regardless of dispersion (§9). A shared default would bury the 622 High-severity inventory anomalies worth reviewing.

11. Limitations

- **Simulated operations.** Stock levels, stockouts, fill rate, unit costs, purchase orders and returns are simulated from real-data distributions and are not validated against real operations. R1's effect sizes are properties of the modelled policy. The *direction* (reorder points must cover lead time) is general inventory theory; the *magnitudes* need confirming on real data.
- **Relabelled geography and suppliers.** Emirates are hashed from source locations, and "suppliers" are DataCo departments. Neither the regional mix nor supplier identities describe a real UAE business.
- **Mixed source periods.** DataCo and Olist cover different windows, so multi-year trends are unreliable (§5).
- **Model scale.** Forecasts use about 2.6 years of weekly history for 8 products. Supplier scores are relative to 11 suppliers, some with very few POs. Anomaly detection ran as one historical pass, not incrementally.
- **Scenario engine.** It uses a single calibrated normal approximation at network level over a 90-day window. It does not model product-level or warehouse-level effects.

12. Future improvements

1. **Validate on real data.** Connect a real inventory and purchase-order feed and re-run R1's policy comparison. The simulator and KPI definitions are already parameterised for it.
2. **Product-level reorder optimisation.** Extend R1 from a uniform "+3 days" to per-product service-level targets using demand variability, not just lead time.
3. **Forecast interval recalibration** (e.g. conformal prediction) so the most accurate point model also has trustworthy intervals.
4. **Production monitoring.** Incremental anomaly detection with per-domain thresholds, plus forecast drift monitoring and scheduled retraining (out of scope for this engagement).
5. **Supplier risk with an outcome label.** Once real supplier-caused disruptions are recorded, benchmark the weighted score against a supervised model.

6. **Complete the external context.** Load UAE CPI and M5 data as originally planned, to test price sensitivity and improve seasonal features.

Reproduce: `python reports/executive_report/build_report.py`. KPIs, figures, lead-time and policy evidence, supplier scores, forecast metrics, anomaly counts and scenario values are recomputed from the warehouse extract and committed phase outputs (`reports/`). Statistical test results (Mann-Whitney, Kruskal-Wallis, Spearman on discounts) and supplier feature values are quoted from the phase notebooks named in each section.